Dartagnan Birnie

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Alton, NH · Available for onsite work in NH and MA

SUMMARY

Software engineer and implementation consultant with a B.S. in Computer Science, deploying and integrating enterprise .NET and SQL Server software for heavy civil construction customers at Trimble. Builds REST and telematics integrations against documented standards, the developer documentation and tools behind them, and full-stack web and desktop apps. Seeking software engineering, implementation, integration, or solutions architect roles.

TECHNICAL SKILLS

Languages: C#, SQL, Python, JavaScript / TypeScript, Java, C, PHP, HTML5 / CSS
Platforms & Data: .NET Framework, .NET 8, Microsoft SQL Server, SSRS, SSMS, IIS, Node.js, Express, React, Electron, PostgreSQL, MongoDB
Integration & APIs: REST APIs, Postman, OAuth 2.0 / JWT, AEMP2 (ISO 15143-3) telematics, Socket.IO, Task Scheduler, batch scripting
Engineering Practice: Requirements capture and BRDs, solution configuration, defect lifecycle (JIRA, Salesforce), root-cause analysis, technical documentation, Agile / Scrum, Git / GitHub, deployment, upgrades, and environment configuration

EXPERIENCE

Trimble Inc. — B2W Software — Portsmouth, NH

Software Implementation Consultant — *May 2026 – Present*

- Lead end-to-end deployments of enterprise construction-estimating software built on the Microsoft .NET Framework and SQL Server: requirements capture, database build-out and data-integrity analysis, configuration, integration, and customer acceptance.
- Build and validate machine-telematics integrations against the AEMP2 (ISO 15143-3) standard, connecting John Deere, Komatsu, and Samsara equipment data to customer systems through REST APIs, scheduled jobs, and batch automation.
- Built and launched b2w-api.com, the developer guide to B2W's three cloud APIs used by customers, partners, and B2W Support: guides with Postman, PowerShell, and cURL examples, plus in-browser tools that check environments, decode API errors, build requests, and inspect JWTs.
- Built FieldLogger, an Electron desktop app for implementation consultants that logs client work in one click, tracks implementation milestones, pulls meetings from Google Calendar, and turns the day into copy-ready time-entry notes, with automatic backups and self-updating installs.
- Serve as the primary escalation point for licensing, activation, and provisioning across the B2W suite, authoring the documentation and workflow improvements that standardized the team's escalation process.

Technical Support Engineer — *Sept. 2025 – May 2026*

- Resolved 1,400+ production issues in six months across SQL Server, authentication, licensing, SSRS, IIS, cloud-hosted infrastructure, and third-party integrations, including installations, upgrades, and migrations.
- Owned the defect lifecycle end to end: reproduced customer-reported failures in test environments, isolated root cause from application logs and SQL Server data, tracked defects in JIRA and Salesforce, and verified fixes with Development and QA before release.
- Promoted to Software Implementation Consultant within six months.

Account Tree — Dover, NH

Software Development & Marketing Intern — *May 2024 – Aug. 2024*

- Developed features and debugged PHP/JavaScript defects in a K–12 activity-fund platform in an Agile cycle, wrote business requirements documents, and rebuilt the company website for its rebrand; featured intern in UNH's 2024 Paid Student Internship at Start-Ups Program.

Designs By Dart LLC — Alton, NH

Founder / Web Developer — *Jan. 2022 – Present*

- Build and maintain sites for 20+ small-business clients; one of two web contractors approved by the NH SBDC and the State of New Hampshire.
- Built a serverless integration for NESTA (nestanh.com): Netlify Functions call the Acuity Scheduling API server-side and feed live class listings into the client's Squarespace site, keeping credentials off the page.

PROJECTS

dartbirnie.dev — Portfolio & Integration Platform — Personal Project

TypeScript, React, Node.js / Express, WebGL (Three.js), Railway

- Built and deployed a portfolio whose Express back end integrates the Spotify (OAuth 2.0 refresh-token flow), Steam, PlayStation, and GitHub APIs behind cached server routes, with a WebGL scene, a scroll-driven timeline, and a site-wide search console.

PiRail — Railroad Track Inspection Platform (UNH Senior Capstone) — Aug. 2024 – May 2025

Software Engineer · React, JavaScript, Python, Raspberry Pi, GPS / IMU / LIDAR

- Delivered the operator interface for a low-cost track-inspection system on a Raspberry Pi with GPS, IMU, and LIDAR sensors, replacing a legacy jQuery interface with a mobile-first React app that plots live position and logs track points of interest.
- Built a playback simulator that replays recorded sensor runs to validate the interface without time on the rail, and coordinated a five-person team across two semesters with Git and Agile practices.

DartERP — Manufacturing ERP — Personal Project

C#, .NET 8, WinForms, Entity Framework Core, SQL Server

- Built a desktop manufacturing ERP modeled on regulated-manufacturing traceability: inventory, purchase orders, work orders, and serialized finished-goods tracking with full audit trails, on a layered .NET architecture over a SQL Server schema and data-access layer I designed.

EDUCATION

University of New Hampshire — *B.S. in Computer Science · Dean's List: Fall 2024, Spring 2025* — **Durham, NH · May 2025**